

Question 1 (12 Marks)

Marks

- a) Solve for x : $3^{2x} - 10(3^x) + 9 = 0$. **3**
- b) A function $f(x)$ is defined as follows.
- $$f(x) = \begin{cases} 0 & \text{for } x < -2 \\ -2 & \text{for } -2 \leq x < 0 \\ x+1 & \text{for } x \geq 0 \end{cases}$$
- i) Evaluate $f(-1) + f(1)$. **1**
- ii) Sketch the function. **3**
- c) Solve the following for x .
- i) $(x-3)(5-x) < 0$ **2**
- ii) $\frac{x}{x-1} \geq 1$ **3**

Question 2 (12 Marks) *Start a new page***Marks**

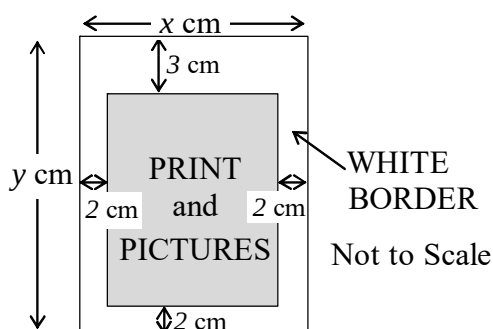
- a) A parabola has equation $y^2 = 8x + 2y + 7$. **6**
- i) Express this equation in the form $(y - y_1)^2 = 4a(x - x_1)$.
- ii) Draw a neat sketch of the parabola indicating the:
- α . co-ordinates of the focus;
 - β . equation of the axis of symmetry;
 - γ . co-ordinates of all points of intersection of the parabola with the co-ordinate axes.
- b) The point $P(-3, 8)$ divides the interval AB externally in the ratio $k : 1$. If A is the point $(6, -4)$ and B is the point $(0, 4)$, find the value of k . **3**
- c) If $\operatorname{cosec} \theta = -\frac{13}{5}$ and $\cos \theta > 0$, find exact values of $\tan \theta$ and $\cos \theta$. **3**

Question 3 (12 Marks) ***Start a new page***

- a) Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 1}{2x^2 + x - 1}$. **2**
- b) Differentiate $4x(5x^2 - 4)^3$ with respect to x leaving your answer in factorized form. **3**
- c) The function $p(x)$ is defined by the rule $p(x) = (x - 1)(x^2 - 5)$.
- i) Find the real roots of the equation $p(x) = 0$.
- ii) Find the coordinates of the turning points of $p(x)$, and determine whether they are maxima or minima.
- iii) Draw a sketch of the graph $y = p(x)$, in the domain $-3 \leq x \leq 3$. **7**

Question 4 (12 Marks) **Start a new page**

- a) Sketch the function given by $f(x) = \frac{x}{|x|}$ 3
- b) Shade the region in the cartesian plane which satisfies $y \leq \sqrt{4 - x^2}$ and $y \geq x^2 - 4$. 3
- c) A children's picture book is being designed so that each page contains 320 square centimetres of print and pictures surrounded completely by a white border as illustrated in the figure below.



Each page is to have a border of width 2 centimetres at the bottom and on each side, as well as a border of width 3 centimetres at the top. Let the width of a page be x centimetres and its length be y centimetres.

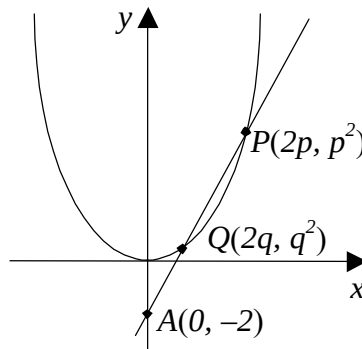
6

- i) Show that the area A square centimetres of one such page is given by
- $$A = x\left(5 + \frac{320}{(x-4)}\right)$$
- ii) Show that the page which fulfills all the printing requirements and which has the smallest area is 20 centimetres wide and 25 centimetres long.

Question 5 (12 Marks) *Start a new page***Marks**

- a) Determine the equation of the locus of the points $P(x, y)$ equidistant from the y -axis and the point $(1, 0)$. **3**

b)

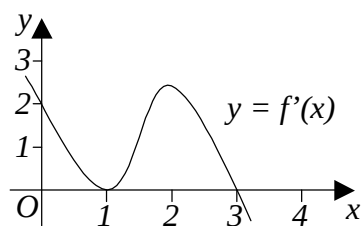


- i) Find the equations of the normals to the parabola $\begin{cases} x = 2t \\ y = t^2 \end{cases}$ at the points $P(2p, p^2)$ and $Q(2q, q^2)$, where $p \neq q$. Hence show that these normals intersect at the point $R(X, Y)$ where $X = -pq(p + q)$ and $Y = (p + q)^2 - pq + 2$.
- ii) If the chord PQ has gradient m and passes through the point $A(0, -2)$ find, in terms of m , the equation of PQ . Hence show that p and q are the roots of the equation $t^2 - 2mt + 2 = 0$.
- iii) By considering the sum and the product of the roots of this quadratic equation show that the point R lies on the original parabola.
- iv) Find the least value of m^2 for which p and q are real. Hence find the set of possible values of the y co-ordinate of R .

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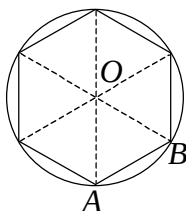
Question 6 (12 Marks) **Start a new page**

a)

**3**

The above diagram shows a sketch of the gradient function of the curve $y = f(x)$. In your Writing Booklet, draw a sketch of the function $y = f(x)$ given that $f(0) = 0$.

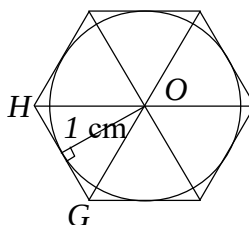
b)



A regular hexagon is drawn inside a circle with centre O so that its vertices lie on the circumference, as shown in the diagram. The circle has radius 1 cm.

- i) Prove that $\triangle OAB$ is equilateral. **2**
- ii) Find the area of $\triangle OAB$ and hence find the area of this hexagon. Leave your answer in surd form. **2**

Another regular hexagon is drawn outside the circle, as shown. The altitude of $\triangle OGH$ is 1 cm.



- iii) Find the area of $\triangle OGH$ and hence find the area of this outer hexagon. Leave your answer in surd form. **3**
- iv) By considering the results in (ii) and (iii), explain why $\frac{3\sqrt{3}}{2} < \pi < 2\sqrt{3}$. **2**

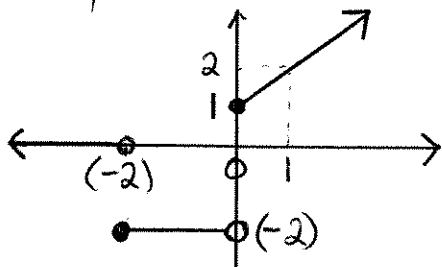
END OF PAPER

YR 11 EXT ONE PRELIM 2006

① a) $3^{2x} - 10(3^x) + 9 = 0$ Let $u = 3^x$
 $u^2 - 10u + 9 = 0$ ①
 $(u-9)(u-1) = 0$
 $u = 1$ or 9 . ① $3^x = 1$ or $3^x = 9$.
 $x = 0$ or $x = 2$ ①

b) (i) $f(-1) + f(1) = -2 + (1+1) = 0$ ①

(ii)

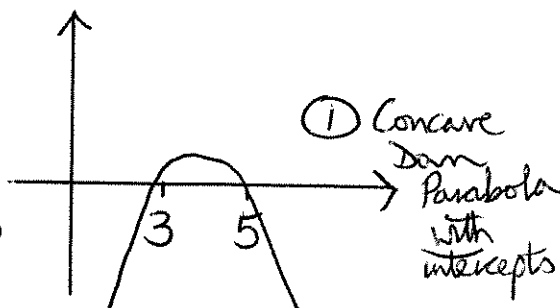


① mark each section including connect endpoints

c) (i) $(x-3)(5-x) < 0$

$x > 5$ or $x < 3$

① Correct Region



① Concave Down Parabola with intercepts

(ii)

$\frac{x}{x-1} \geq 1$

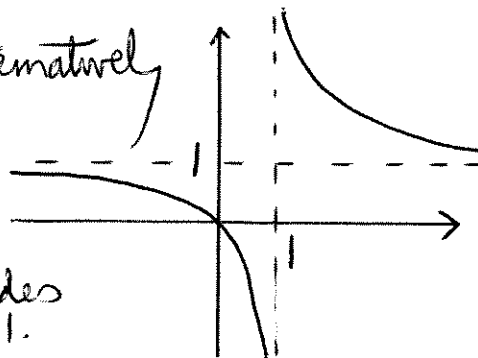
$\frac{x(x-1)^2}{x(x-1)^2} \geq \frac{(x-1)^2}{(x-1)^2}$ ①

$x^2 - x \geq x^2 - 2x + 1$

$x \geq 1$ ①

BUT $x \neq 1$ $\therefore x > 1$ ① Excludes $x = 1$.

Alternatively



MARKERS COMMENT

a) Those recognising an equation reducible to quadratic completed well. Some dropped equals zero making an expression not an equation. Some found u but failed to find x .

b) Lots failed to evaluate $f(1)$ giving answer of $x-1$!

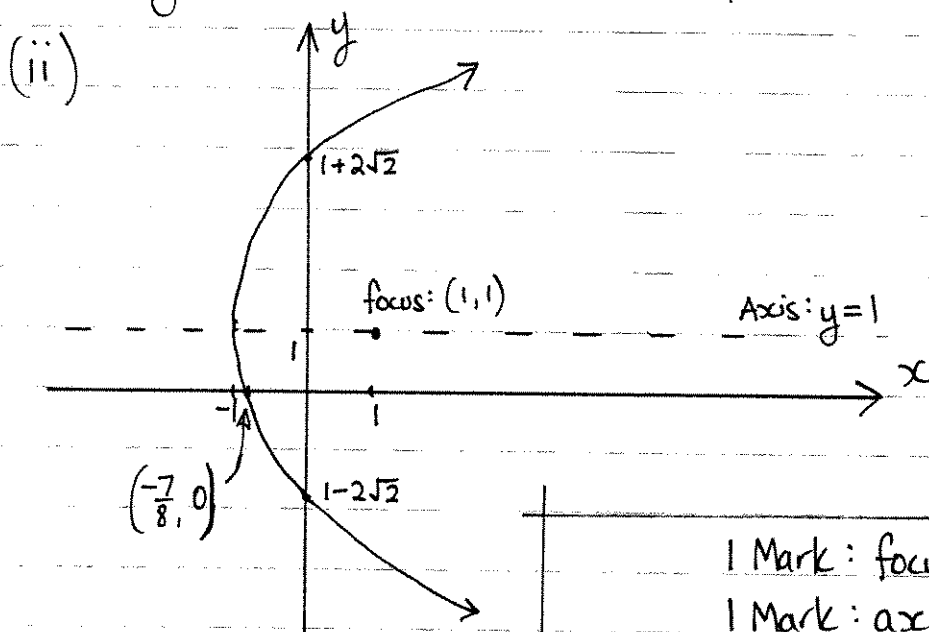
(ii) Common mistakes on graph were failure to deal with endpoints & critical values missing from axes

c) (i) Many drew concave up parabola but then identified correctly the region below x axis. Some confuse two separate regions with writing a single inequality with finite endpoint i.e. $3 > x > 5$!

(ii) Most common mistake was failure to exclude $x = 1$ from solution

Q2 (a)(i) $y^2 = 8x + 2y + 7$
 $\therefore y^2 - 2y = 8x + 7$
 $\therefore (y-1)^2 = 8x + 8$
 $\therefore (y-1)^2 = 8(x+1)$

1 Mark - Completing the square
 1 Mark - final answer



1 Mark: focus
 1 Mark: axis
 1 Mark: y-intercepts
 1 Mark: x-intercept
 (based on answer for part (i))

(b) $x_p = \frac{mx_2 + nx_1}{m+n}$

$\therefore -3 = \frac{k \cdot 0 + 1 \cdot 6}{k+1}$

$\therefore -3k - 3 = 6$

$\therefore -3k = 9$

$\therefore k = -3$

3 Marks : $k = \pm 3$ (either accepted)

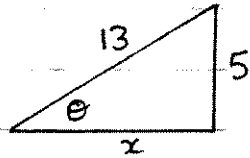
2 Marks : one error (eg. swapped x_1 and x_2)

1 Mark : multiple errors, but tried to use correct method

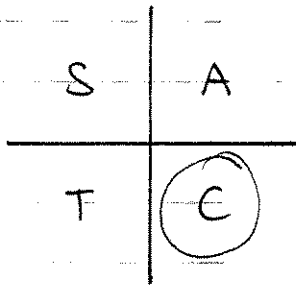
0 Marks : no idea

Q2. (c) $\operatorname{cosec} \theta = -\frac{13}{5}$, $\cos \theta > 0$

$\therefore \sin \theta = -\frac{5}{13}$



$x^2 = 13^2 - 5^2$
 $x = 12$



$\sin \theta < 0$
 $\cos \theta > 0$
 $\therefore 4^{\text{th}} \text{ quadrant}$

$\tan \theta = -\frac{5}{12}$

$\cos \theta = \frac{12}{13}$

1 Mark
 - Quadrant

1 Mark

1 Mark

Q3 Y11 Prelim 06 Ext 1

(a) $\lim_{x \rightarrow \infty} \frac{3 - \frac{2}{x} + \frac{1}{x^2}}{2 + \frac{1}{x} - \frac{1}{x^2}} = \frac{3}{2}$ ✓ (1)

(lim notation poor
 $\lim_{x \rightarrow \infty} =$ deduction of mark
 $\frac{3}{2}$ no working 1 mark)

(b) $\frac{d}{dx} 4x(5x^2-4)^3 = 4x \times 3(5x^2-4)^2 \times 10x + 4(5x^2-4)^3$
 $= 4(5x^2-4)^2 (30x^2 + 5x^2 - 4)$
 $= 4(5x^2-4)^2 (35x^2 - 4)$ (3)

product rule (1)
 1 for diff.

1 for factorization

(no product rule no marks)

(c) (i) $(x-1)(x^2-5)=0$
 $x=1, \pm\sqrt{5}$ ✓

still many $\sqrt{5}$!
 excluding

(ii) $p'(x) = (x-1)2x + (x^2-5)$
 $= 2x^2 - 2x + x^2 - 5$
 $= 3x^2 - 2x - 5$
 $= (3x-5)(x+1)$ ✓

stat pts when $p'(x)=0$

(i) $(3x-5)(x+1)=0$
 $x=-1, \frac{5}{3}$

$p''(x) = 6x - 2$

At $x=-1$ $p(-1)=8$

$p''(-1) = -8 < 0$

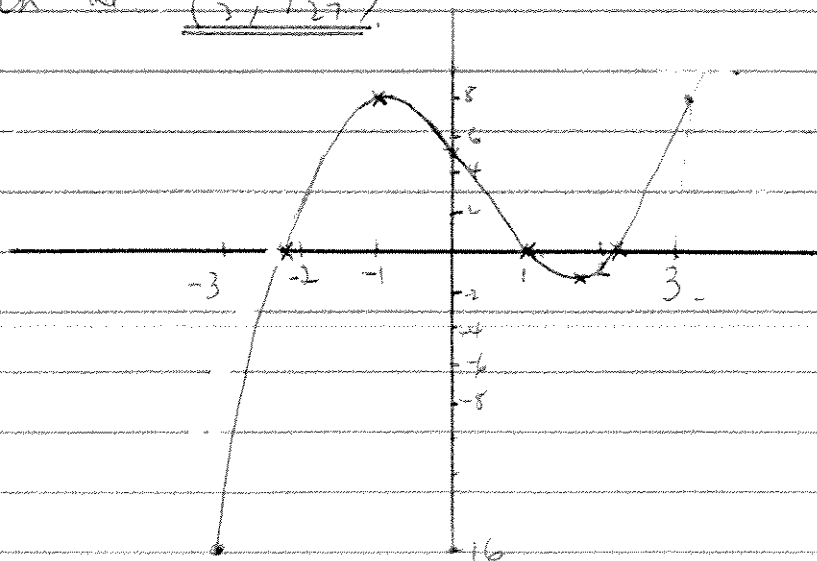
(1 for nature)

$\therefore$ max at $(-1, 8)$ ✓

At $x=\frac{5}{3}$ $p(\frac{5}{3}) = -1\frac{13}{27}$

$p''(\frac{5}{3}) = 8 > 0$

$\therefore$ min at $(\frac{5}{3}, -1\frac{13}{27})$ ✓



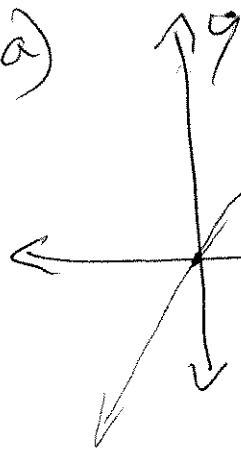
✓ 1 turning pts

1 intercept +

end pts

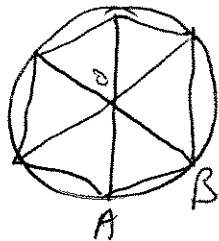
(many marks off extreme values)

Q6 a)



at $x=1$ Pt of inflection -
 at $x=2$ Pt of inflection -
 $y=f(x)$ at $x=3$ MAX turning pt -

(b)



i) Prove that $\triangle OAB$ is equilateral

Method 1 - Geometry

$$\text{Angle at center} = \frac{360}{6} \text{ [regular Hexagon]} \\ = 60^\circ$$

as $\triangle OAB$ is isosceles [$OA=OB$]

$$\angle OAB = \angle OBA \text{ [base angles equal]}$$

$$\therefore 2\angle OBA + 60^\circ = 180^\circ \text{ [sum of } \angle\text{s]}$$

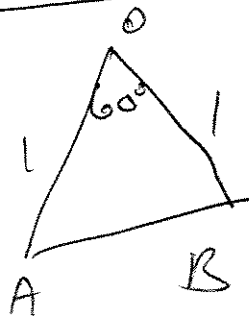
$$\therefore \angle OBA = 60^\circ, \angle OAB = 60^\circ$$

$\therefore \triangle OAB$ is equilateral

Notes:

Reasons must be given.

Method 2 - Trigonometry



$$AB^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \cos 60^\circ$$

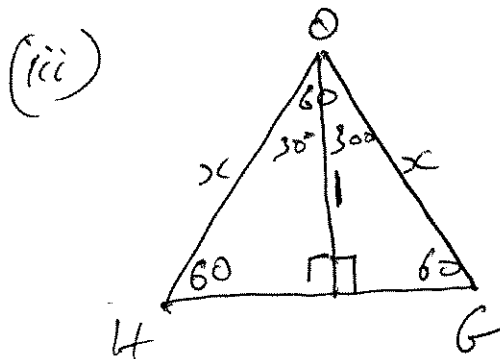
$$= 2 - 2 \times \frac{1}{2}$$

$$= 1$$

$\therefore \triangle OAB$ is equilateral.

$$\begin{aligned}
 (b) (ii) \quad \text{Area of } \triangle AOB &= \frac{1}{2} ab \sin C \\
 &= \frac{1}{2} \times 1 \times 1 \times \sin 60 \\
 &= \frac{\sqrt{3}}{4} \text{ cm}^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Area of hexagon} &= 6 \times \frac{\sqrt{3}}{4} \\
 &= \frac{3\sqrt{3}}{2} \quad \checkmark
 \end{aligned}$$



$$\begin{aligned}
 \frac{1}{x} &= \cot 30^\circ \\
 x &= \frac{1}{\cot 30^\circ} \therefore \text{Area of outer hexagon} \\
 &= \frac{1}{\frac{1}{\sqrt{3}}} \\
 &= \frac{\sqrt{3}}{1} \quad \checkmark \\
 &= \frac{2\sqrt{3}}{3} \quad \checkmark \\
 &= 6 \times \frac{1}{2} \times \frac{2\sqrt{3}}{3} \times \frac{2\sqrt{3}}{3} \times \sin 60^\circ \\
 &= 4 \times \frac{\sqrt{3}}{2} \\
 &= 2\sqrt{3} \text{ cm}^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (iv) \quad \text{Area of Circle} &= \pi r^2 \\
 &= \pi \times 1^2 \\
 &= \pi \quad \checkmark
 \end{aligned}$$

Area of Inner Hexagon < Area of Circle < Area of Outer Hexagon

$$\therefore \frac{3\sqrt{3}}{2} < \pi < 2\sqrt{3} \quad \checkmark \quad \text{from (i) \& (iii)}$$